

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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DAT316 - R

Dive deep into Amazon DynamoDB using design puzzles

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Agenda

We're going to collaboratively explore a set of design puzzlers:

1. "Amuse bouche" puzzlers to get us warmed up
2. Three main entrée deep dives
3. Dessert!

“Amuse bouche” warm-ups

“

I'm seeing 10,000 requests per second but am consuming 20,000 WCUs. Why?

10,000 consuming 20,000

Oh, and my table's average item size is under 1kb, does that change your answer?

Could indexes have something to do with this?

“

I have global tables and the write rates don't match across regions. Why?

Global table writes

Could it have something to do with TTL?

What about a failure or network glitch?

What if you update and don't actually make a change?

Could a condition expression be involved?

“

I want a downstream client to do bulk reads but limited to a certain amount of read capacity.

Can I do this without any throttle risk to other users?

Isolated table scans

Might a global secondary index (GSI) help?

At what cost?

Should its read capacity be auto scaled?

Is there any other use for this GSI design using the same PK + SK?

“

I want to sample 1,000 random items from my table. How?

Pulling a random item

How does a scan work?

How does a parallel scan work?

How many total segments can we use with a parallel scan?

Is there a need to use `limit`?

Bulk scan and write



“

**I want to bulk update (or delete)
a good chunk of items in my
table. Is there an easy way to do
this? No downtime allowed!**

Tooling for bulk update: Amazon EMR or AWS Glue

Could we use Amazon EMR? Or AWS Glue?

[AWS Database Blog](#)

Backfilling an Amazon DynamoDB Time to Live attribute using Amazon EMR: Part 2

by Juhi Patil | on 01 JUL 2022 | in [Amazon DynamoDB](#) | [Permalink](#) | [Comments](#) | [Share](#)

This technique works best for insert-only workloads or during periods of maintenance when updates and deletes aren't allowed. If data is being updated or deleted during the process, it may be unintentionally deleted.

Coding a scan and update

Goal: Work against a live table, use condition expressions, control timing and rate limiting

Assume a single-threaded full table scan. Reads a set of items, updates some of the items. How fast can this run?

What if it updates every item? What if it updates 1% of items?

Does the bottleneck have any negative impact beyond slowing our speed?

Coding a scan and update – In parallel

Okay, let's try a parallel scan with 10 threads. Is there still a bottleneck?

How can we eliminate the bottlenecks?

Coding a scan and update – Shuffle scan

Imagine a parallel scan with X many segments.

Pick a random segment. Process the next Y many items from it. Repeat with a new random segment.



Keep track of your progress within each segment by tracking its LastEvaluatedKey. When the LEK comes back empty, that segment is done.

What's the benefit of this approach?

Optimizing costs



“

(Redacted) would like assistance in ways to estimate and reduce the cost when modifying a single attribute on ~550 billion DynamoDB items

Reserved capacity

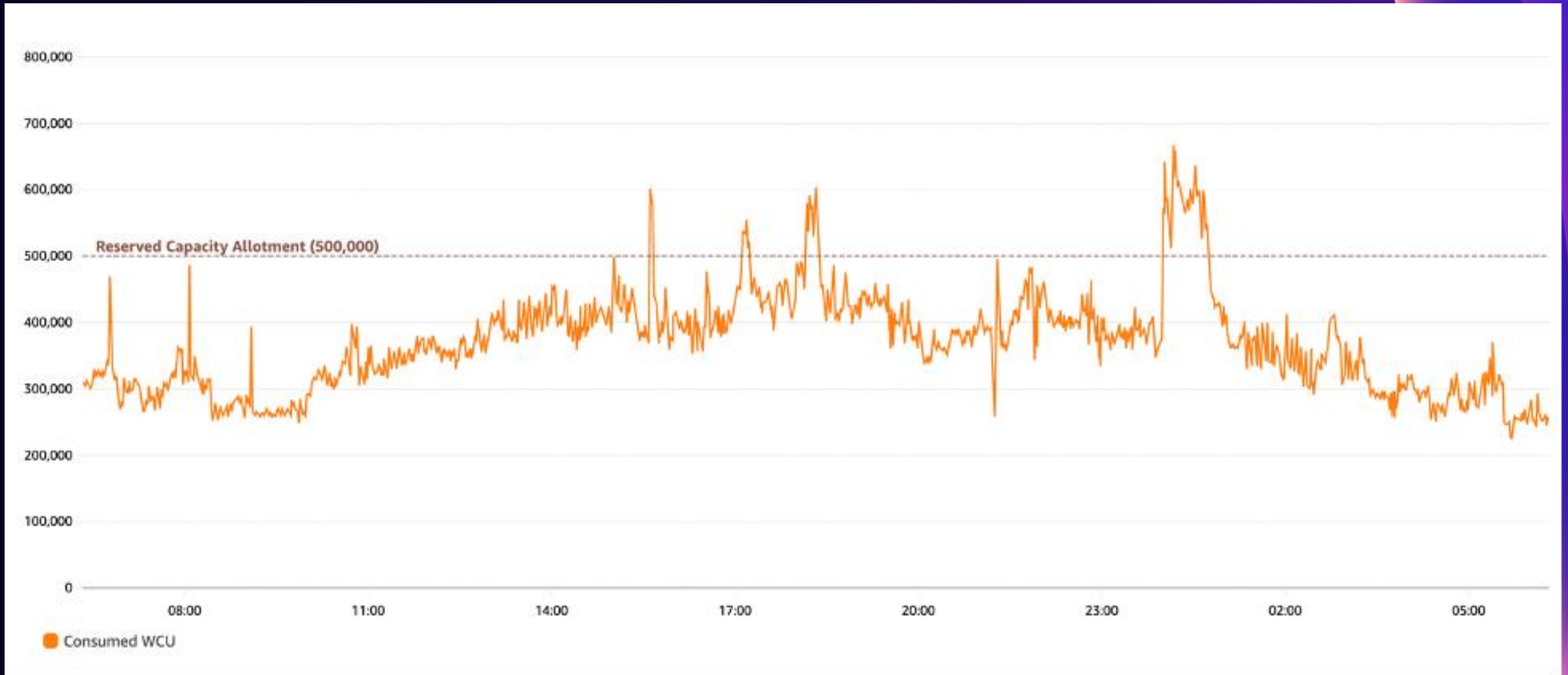
Is there a way to use reserved capacity?

Hint: Reserved capacity, if purchased, is a minimum commitment. It's greatly discounted but is also use it or lose it. Mathematically, the optimal amount to purchase is generally above the daily minimum.



"Reco" tool

Reserved capacity



Reserved capacity challenges

How best to calculate the periods and amount of underutilization?

Remember, RC is across all tables in a region under the payer account!

How best to ensure the bulk process runs at the right consumption rate?

Blog covering three approaches

[AWS Database Blog](#)

Cost-effective bulk processing with Amazon DynamoDB

by Jason Hunter | on 30 MAY 2023 | in [Amazon DynamoDB](#), [Expert \(400\)](#), [Technical How-To](#) | [Permalink](#) | [Comments](#) | [Share](#)

Timing matters



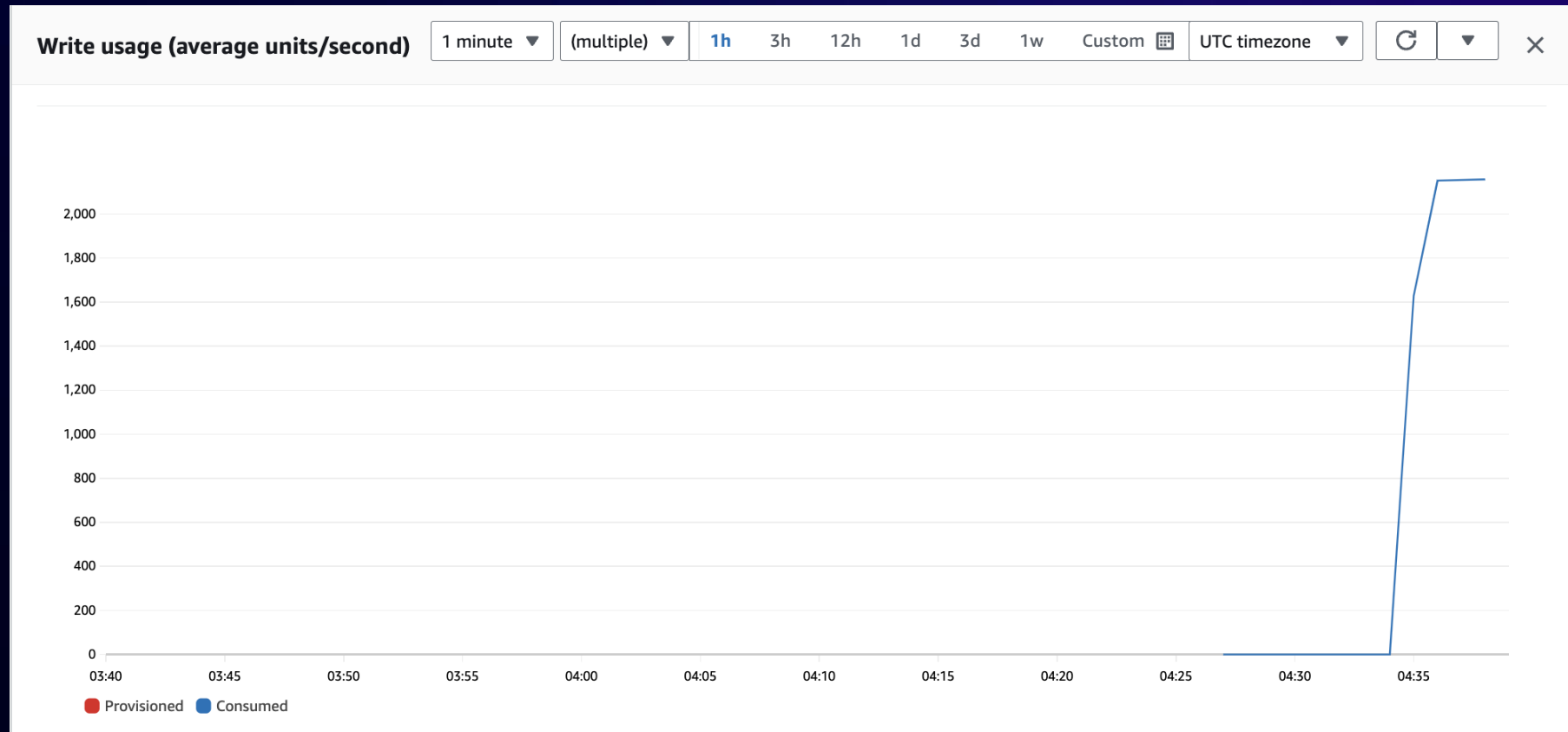
“

**I configured 30,000 WCUs.
I see lots of throttles and
CloudWatch shows I'm
consuming just 2,500 WCUs.**

**Partition key is UUID. No sort
key. Insert only workload. Why?**

CloudWatch proof

- Amazon CloudWatch confirms just 2,500 average units/second



Units and resolution

- What are the units?
- What's the resolution?

Write usage (average units/second)	1 minute ▼	(multiple) ▼	1h	3h	12h
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What 2,500/sec truly means is 150,000/minute.

What if the 150,000 writes come at the top of the minute?

What's the fix?

Dessert



“

**I have PITR on a large table.
Someone ran a bad bulk job
Monday at noon that deleted
important items. How can I
recover?**

PITR

Is there anything better than a full table restore at 11:50 AM?

Hint: Don't do a restore, do an export

Hint: Don't do a full export

Consider an incremental export from 11:30 AM to 12:30 PM that puts on disk the OldImage and NewImage of all modified items!

What's the cost savings if the table is 100 TB?

Incremental export to Amazon S3

☐ Full export

Export the table data in its current state, or from any specific point up to 35 days ago.

☒ Incremental export - *new*

Export any table data that's changed within a specific time period.

Export period [Info](#)

Specify when the incremental export starts and ends. Your earliest export point is August 30, 2023, 00:21:28 (UTC-07:00), as of October 4, 2023, 00:16:28 (UTC-07:00).

 Specify the start and end of export period

Relative mode

Absolute mode



September 2023

October 2023



Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
					1	2	1	2	3	4	5	6	7
3	4	5	6	7	8	9	8	9	10	11	12	13	14
10	11	12	13	14	15	16	15	16	17	18	19	20	21
17	18	19	20	21	22	23	22	23	24	25	26	27	28
24	25	26	27	28	29	30	29	30	31				

Start date

Start time

End date

End time

2023/10/02

11:30:00

2023/10/02

12:30:00

The export period must be between 15 minutes and 24 hours. For date, use YYYY/MM/DD. For time, use 24 hr format.

your KMS key.

Incremental export file format

```
{ "Metadata":  
  { "writeTimestampMicros": "1680109764000000" },  
  "Key": { "PK": { "S": "CUST#200" } },  
  "OldImage": {  
    "PK": { "S": "CUST#200" },  
    "FirstName": { "S": "Mary" },  
    "LastName": { "S": "Grace" } },  
  "NewImage": {  
    "PK": { "S": "CUST#200" },  
    "FirstName": { "S": "Mary" },  
    "LastName": { "S": "Smith" } } }
```

“

**I want cross-Region disaster recovery.
I'm okay with 6 hours RPO and RTO.
Is there a low-cost option?**

Options for disaster recovery

- Global tables?
- Use PITR with cross-Region restore?
- Use AWS Backup to transfer an on-demand backup to a new Region?
- Anything else?

Disaster recovery the incremental way

- Assume a 10 TB table with 1 GB of log accumulation every 4 hours...
 - \$0.10/GB exported → \$0.10 per export * 6 per day = \$0.60/day
 - \$0.02/GB sent → \$0.01 per compressed export * 6 per day = \$0.06/day
 - Plus write costs to the remote table during any DR recovery

Thank you!

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Please complete the session
survey in the mobile app